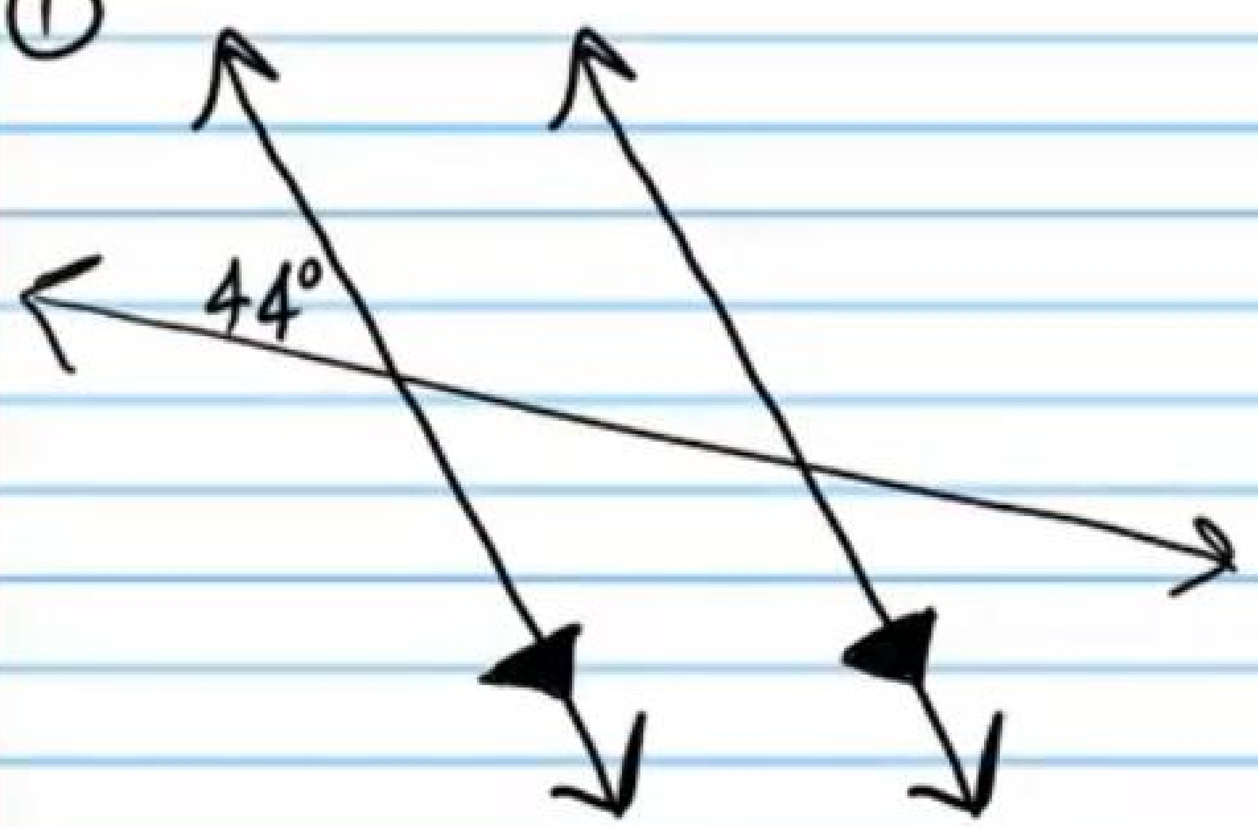
Name
Date
Course

Angles Formed by Transversals (Homework)

Given one angle measure below, find all other angles.

Answers on subsequent pages.

1



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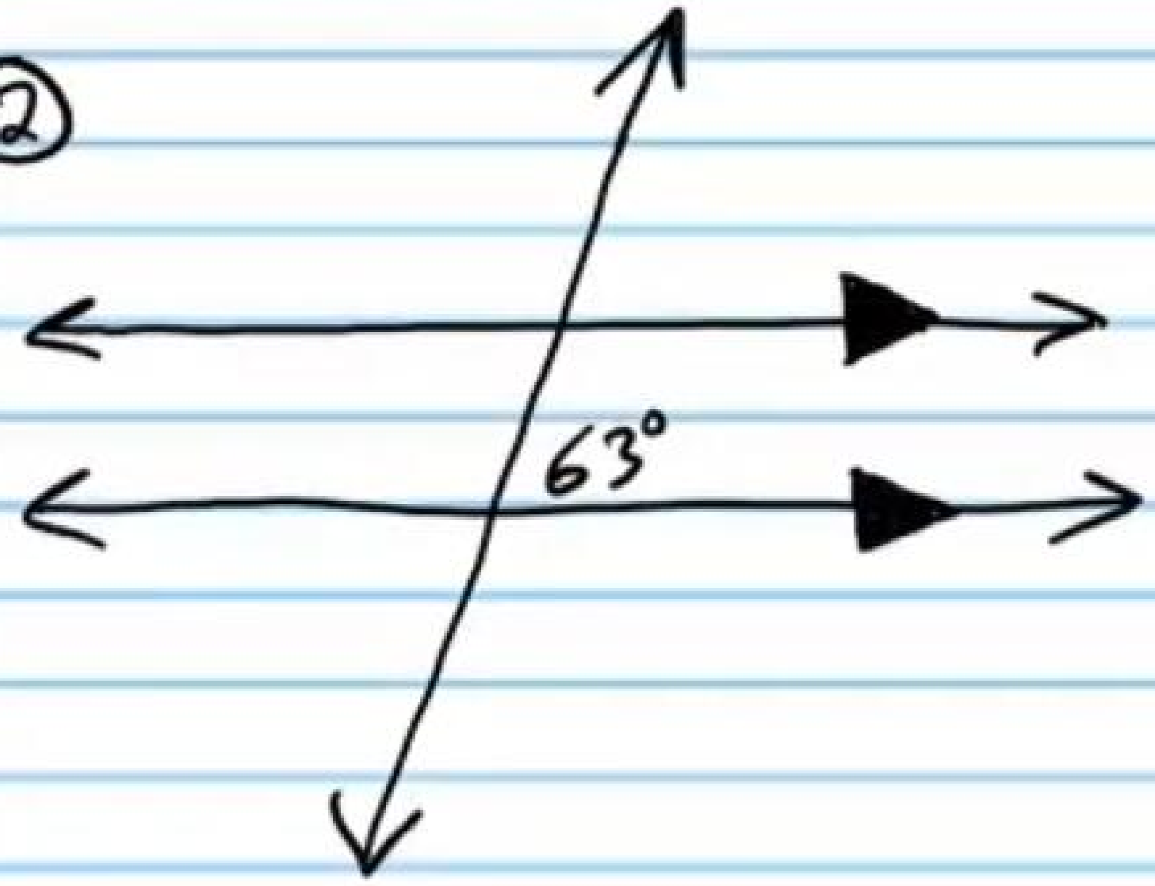

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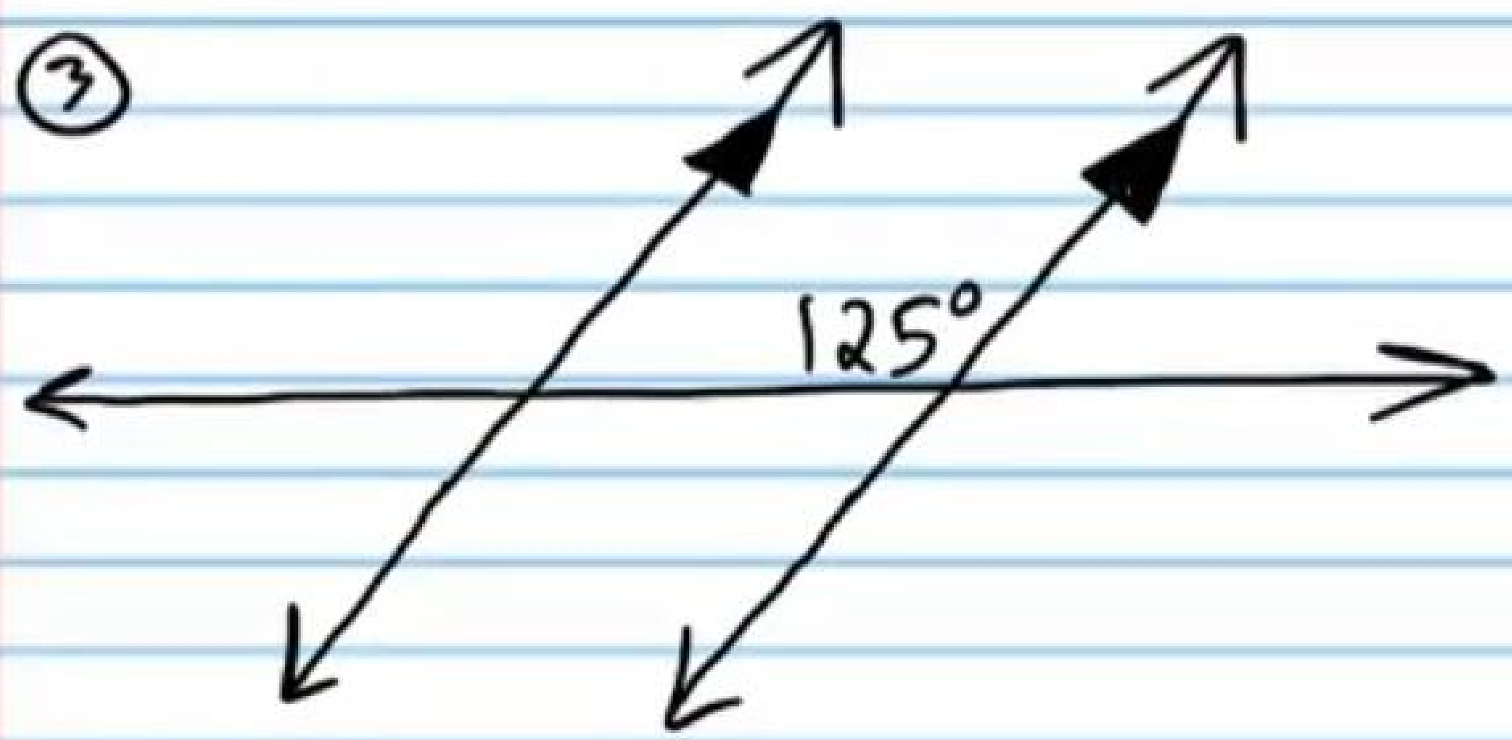

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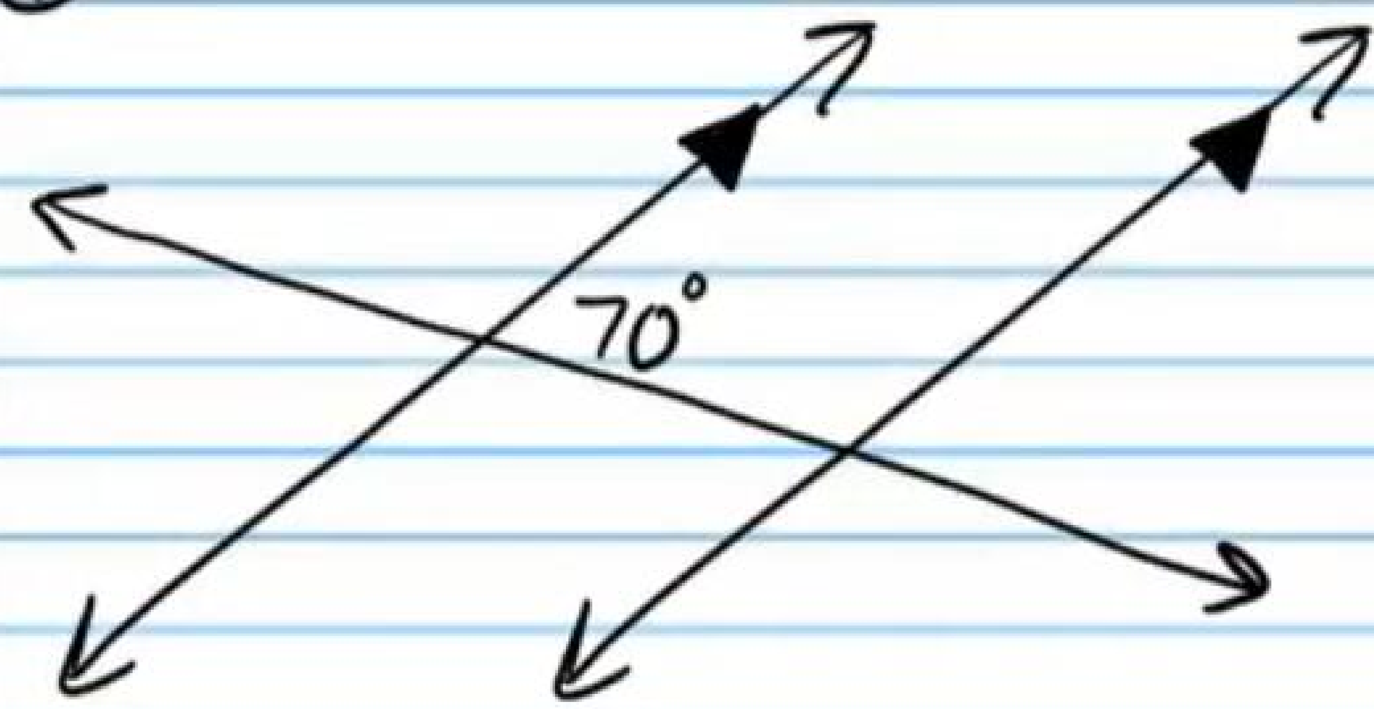

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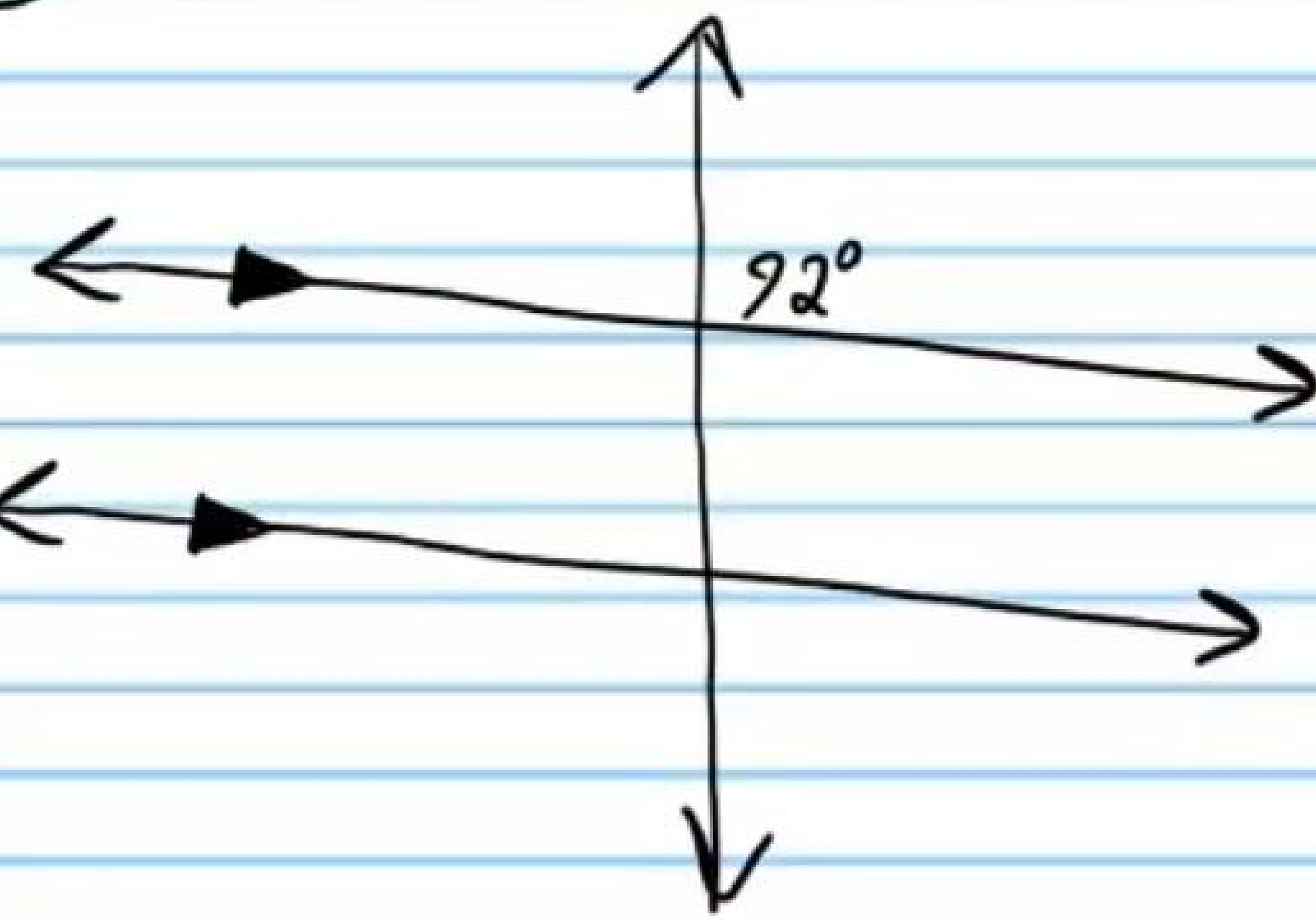

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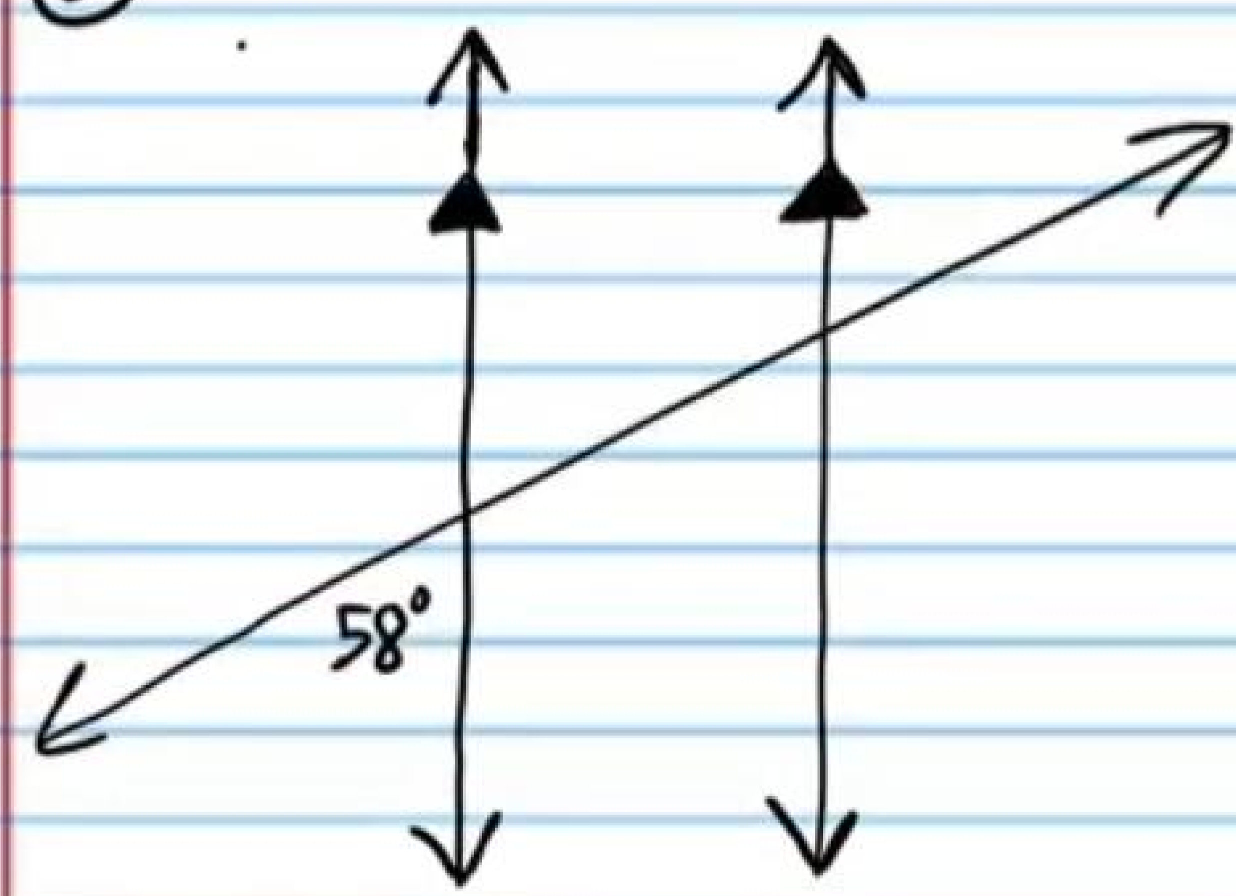
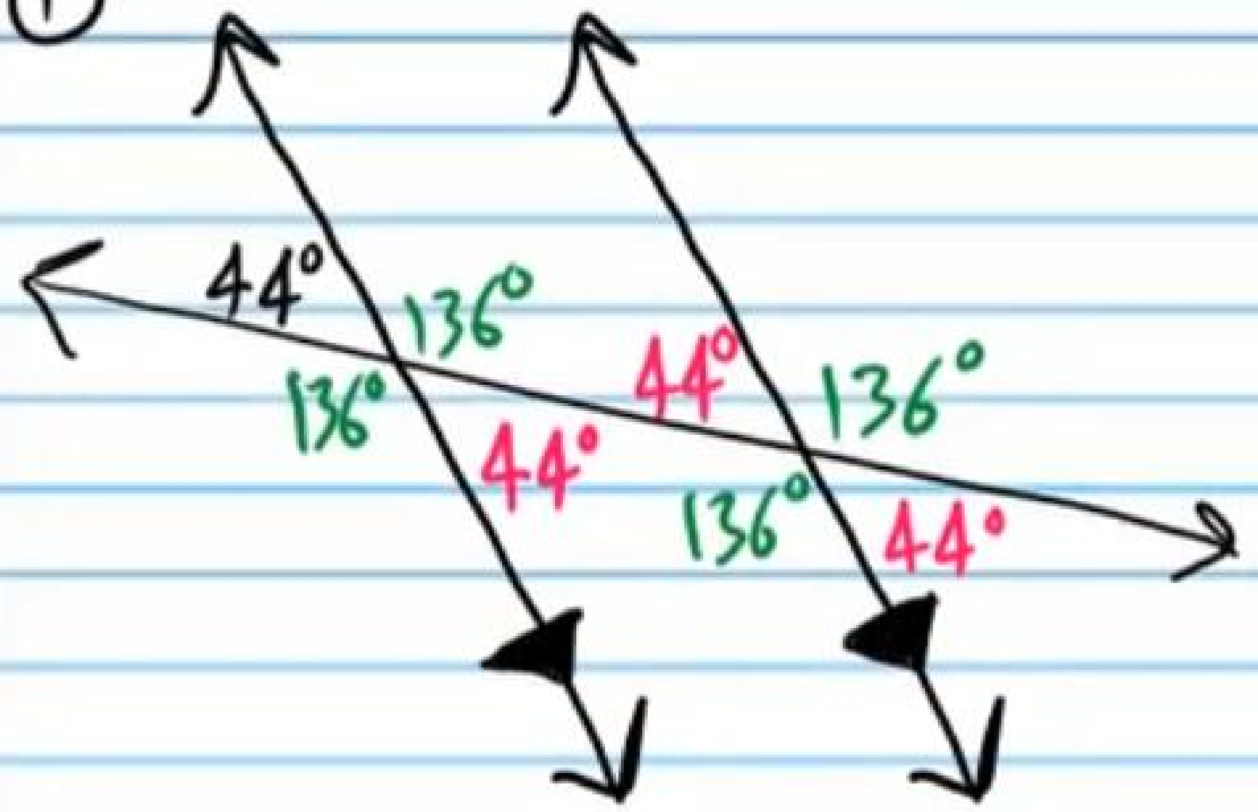


Answers for #'s 1-6.

①



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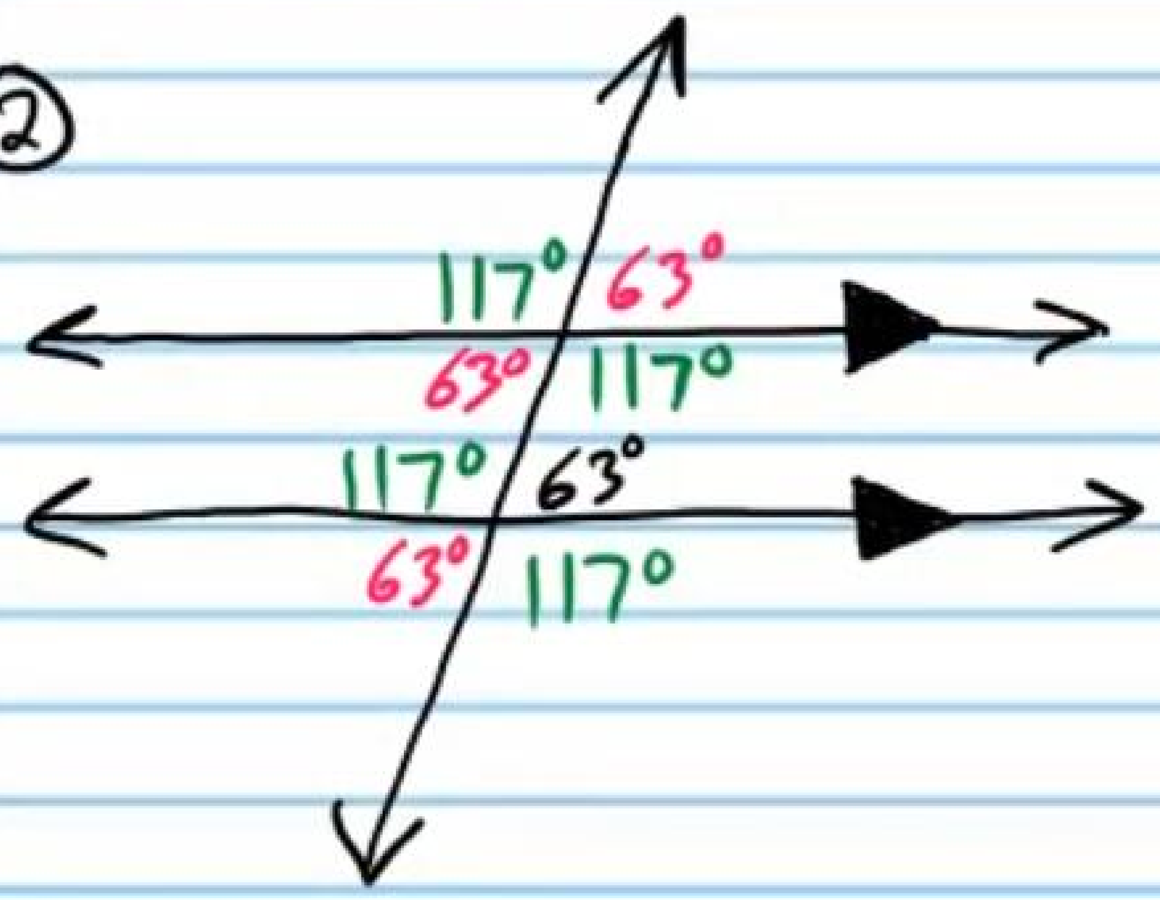
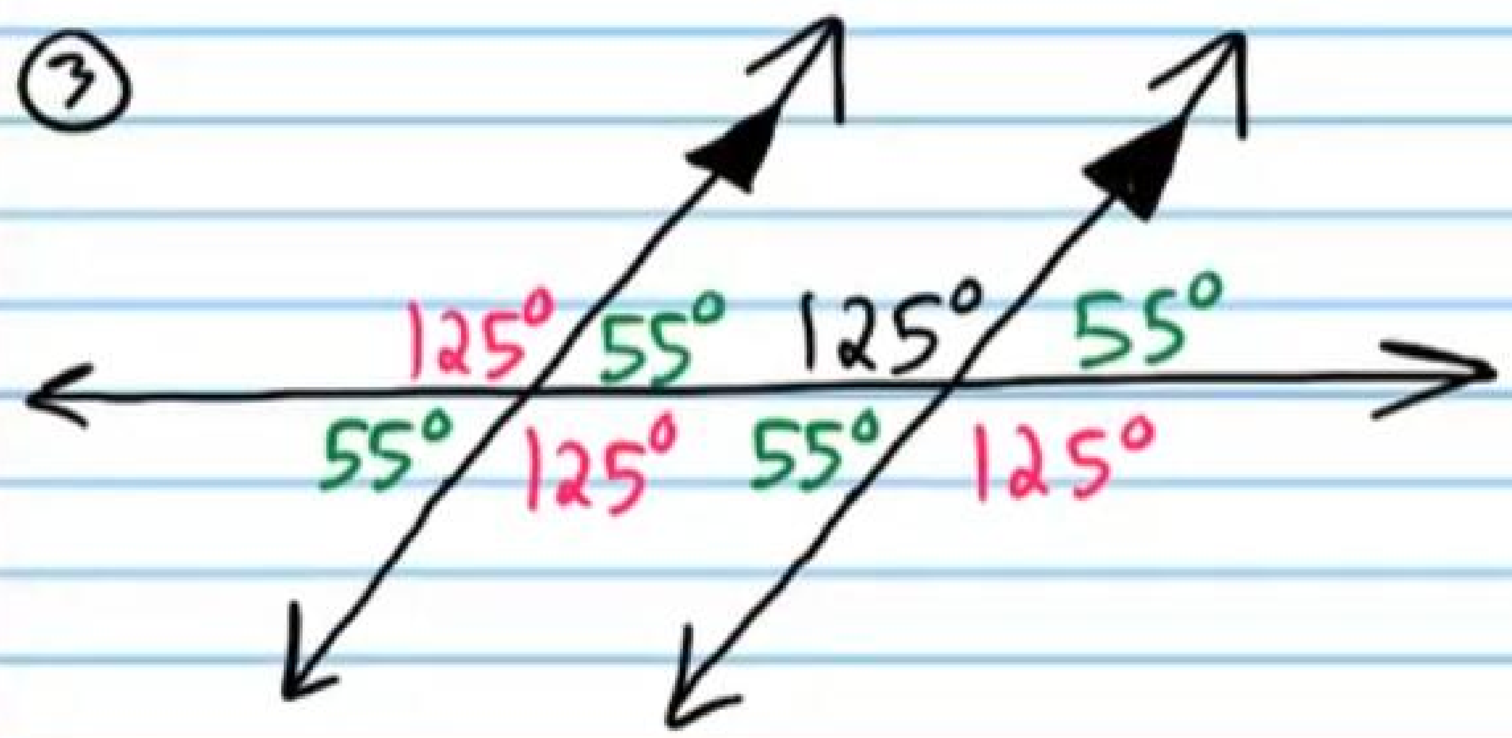
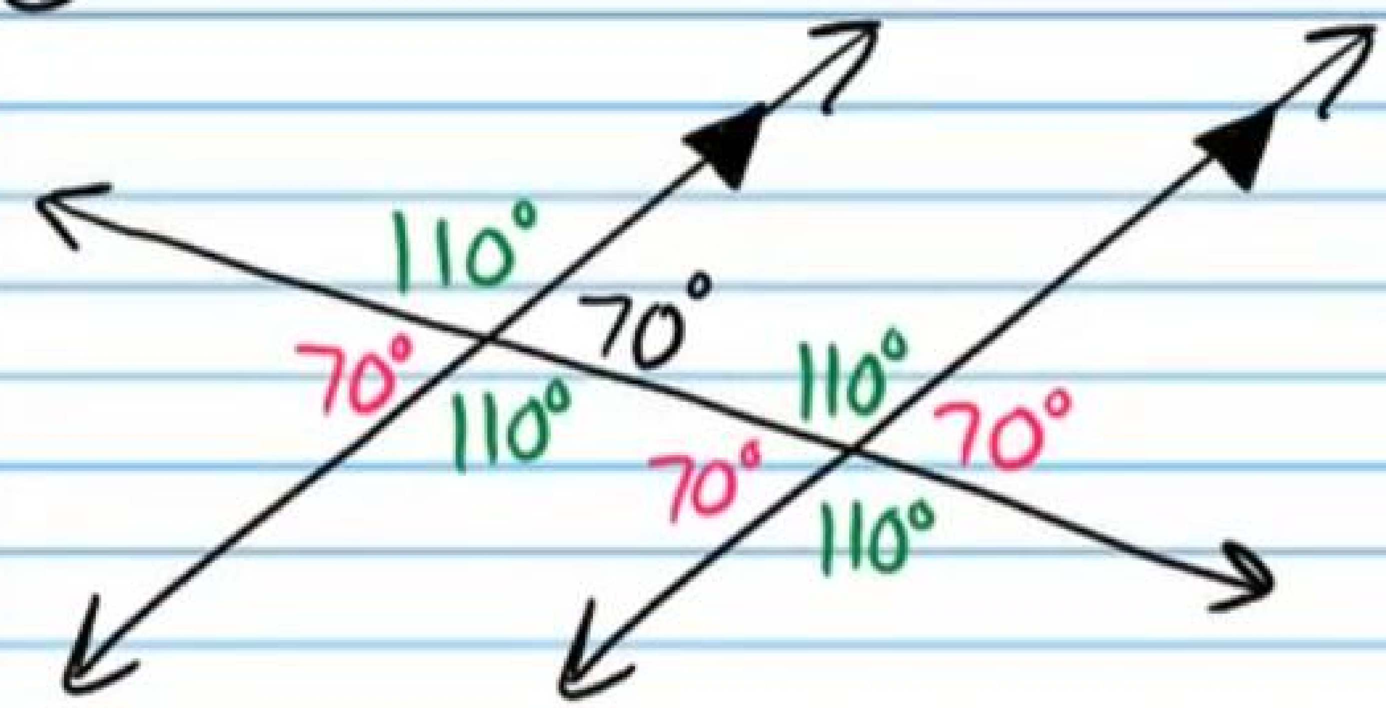


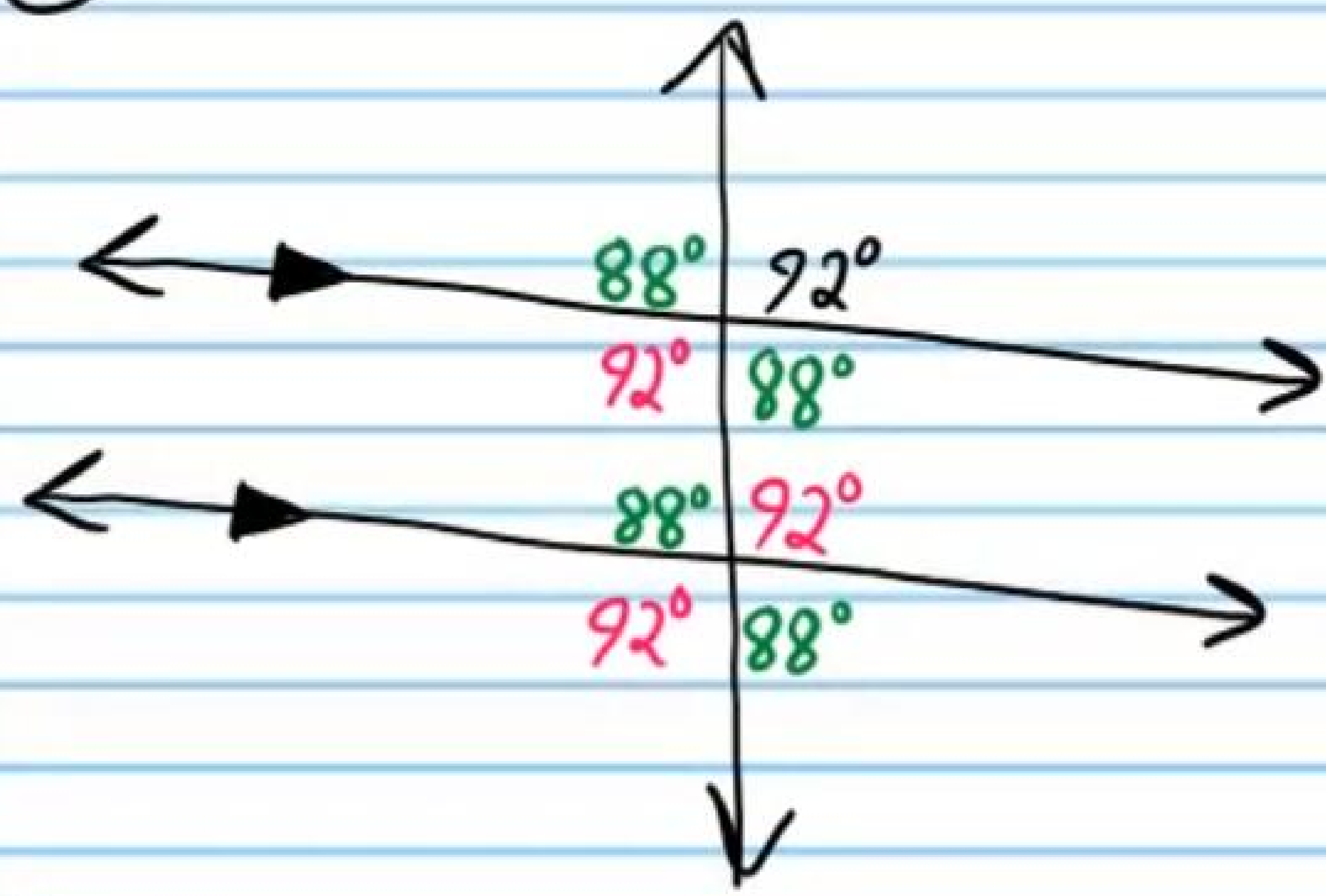
③



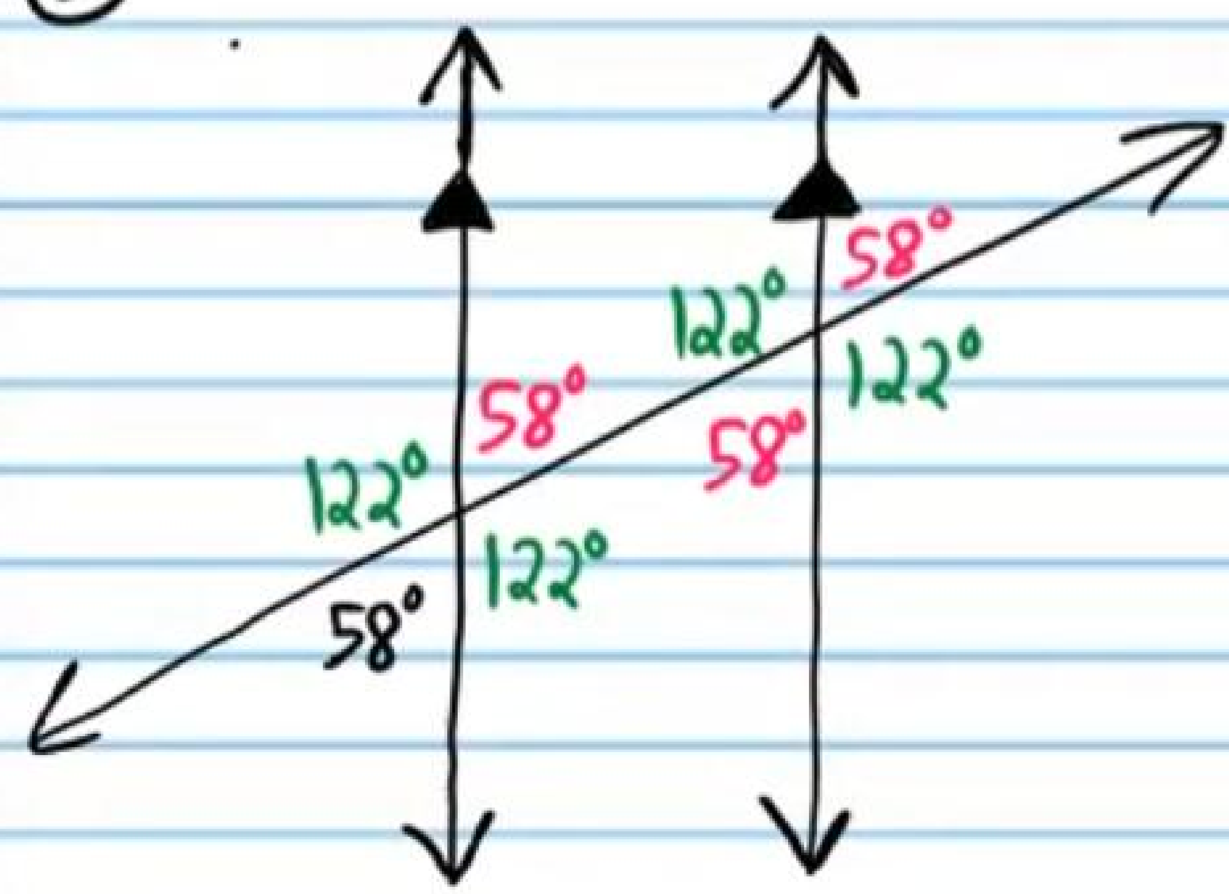
④



⑤



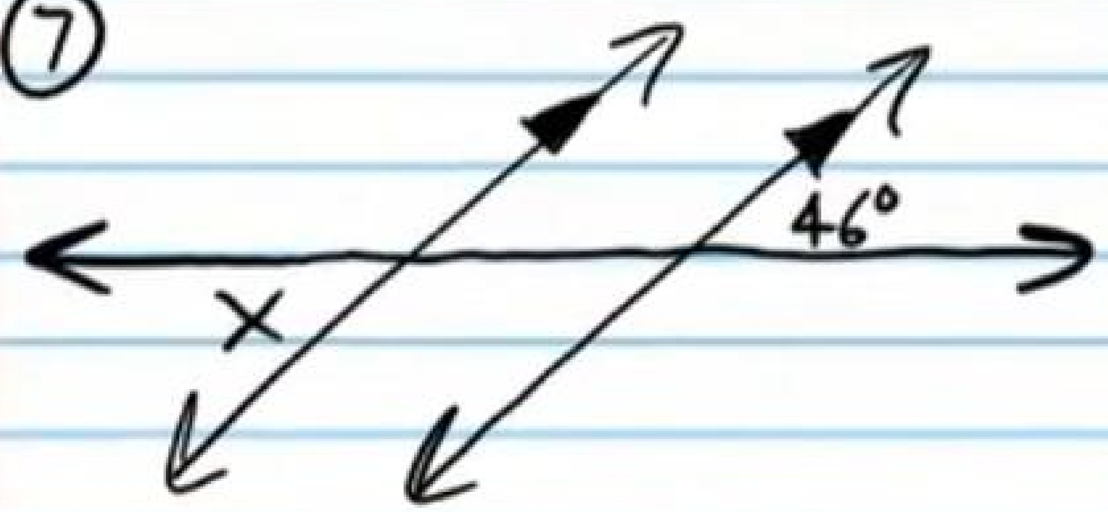
⑥



Find x and state the theorem(s) and/or postulate(s) used.

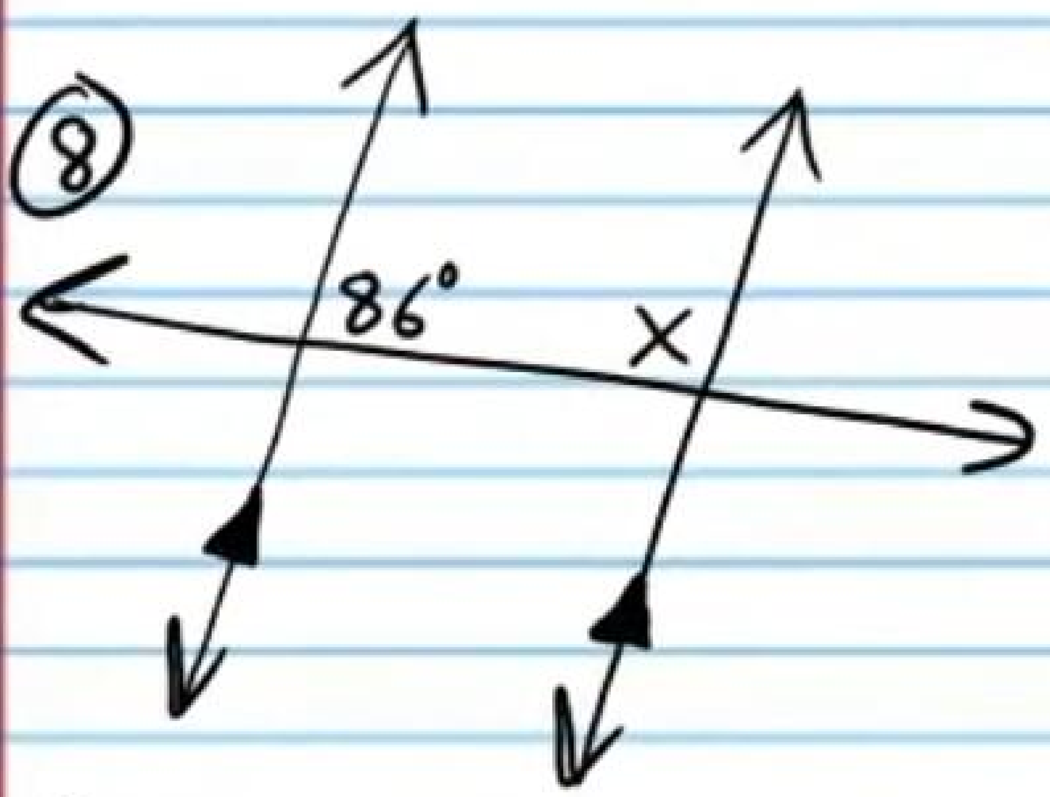
Answers

(7)



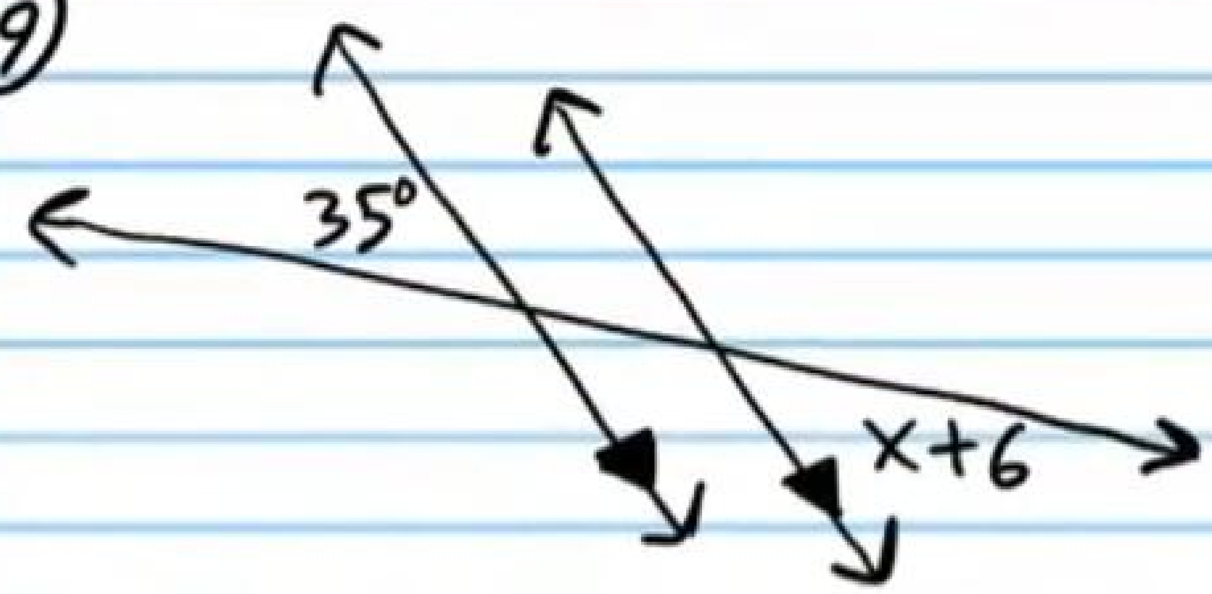
X=46°
AEAT

(8)



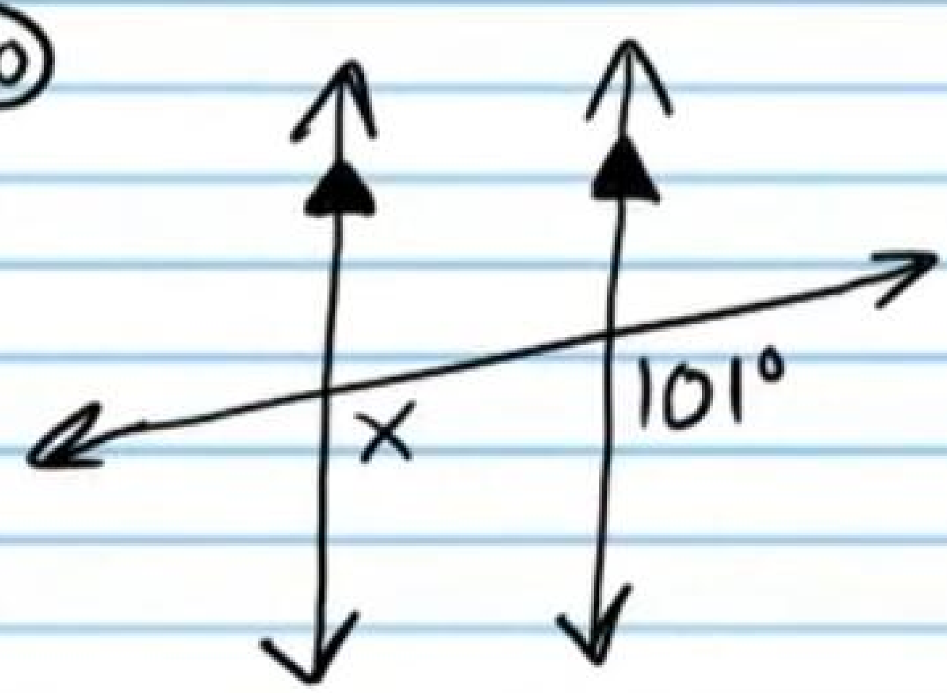
X=94°
CIAT

(9)



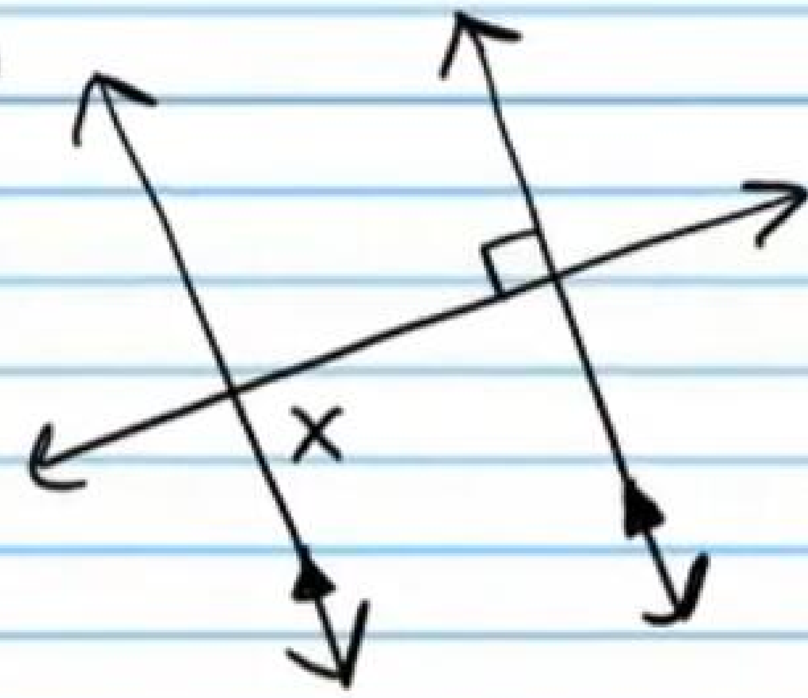
X=29°
AEAT

(10)



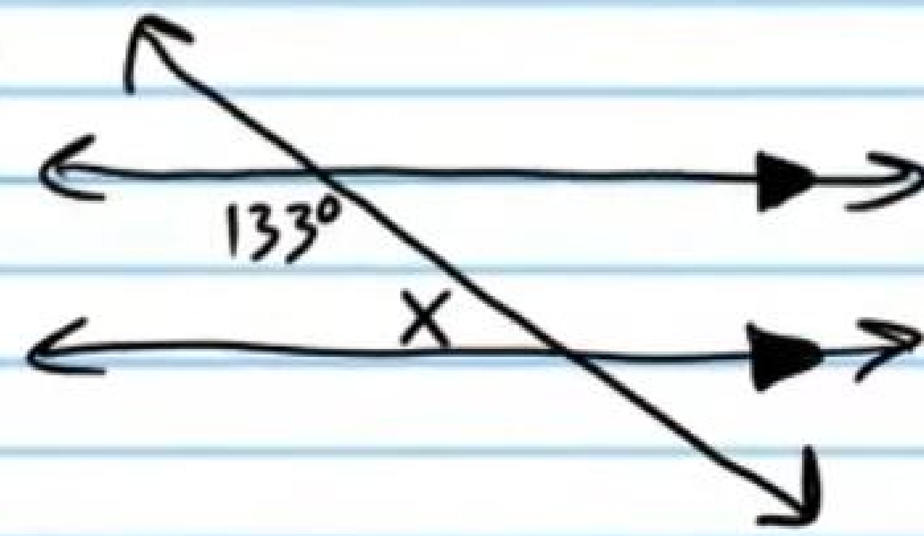
X=101°
CAP

(11)



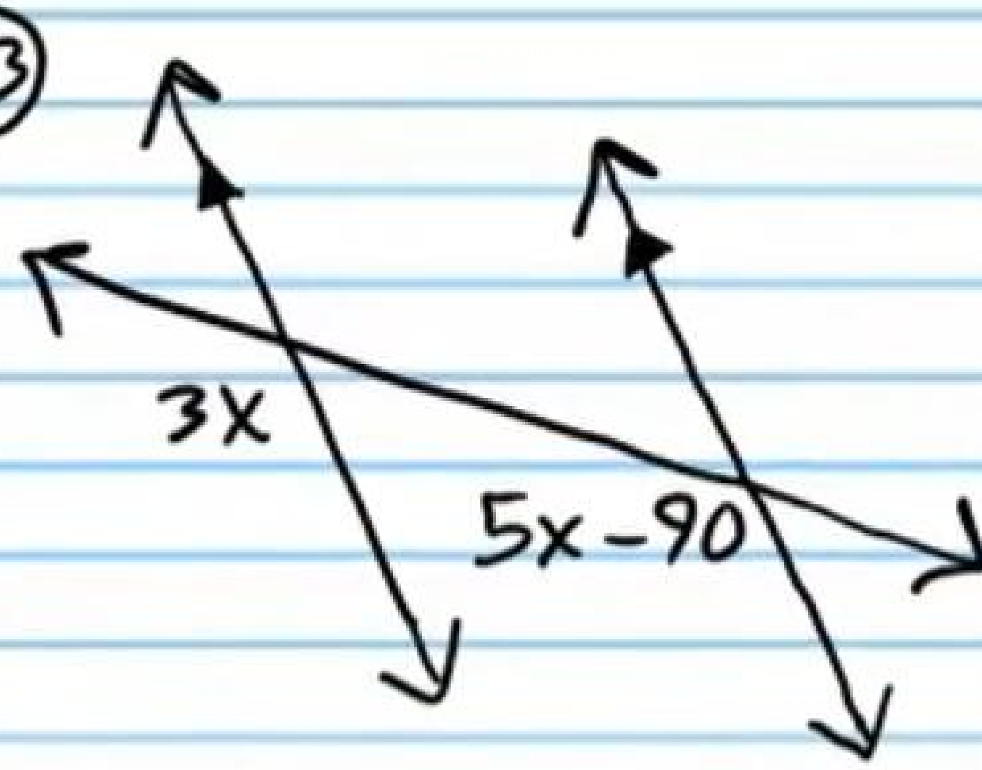
X=90°
AIAT

(12)



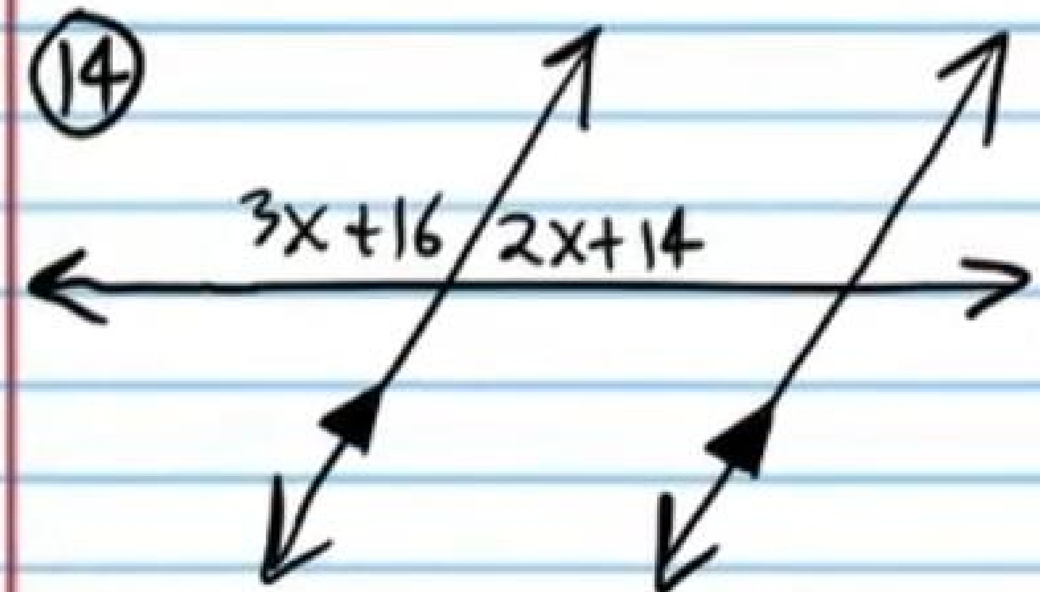
X=47°
CIAT

(13)



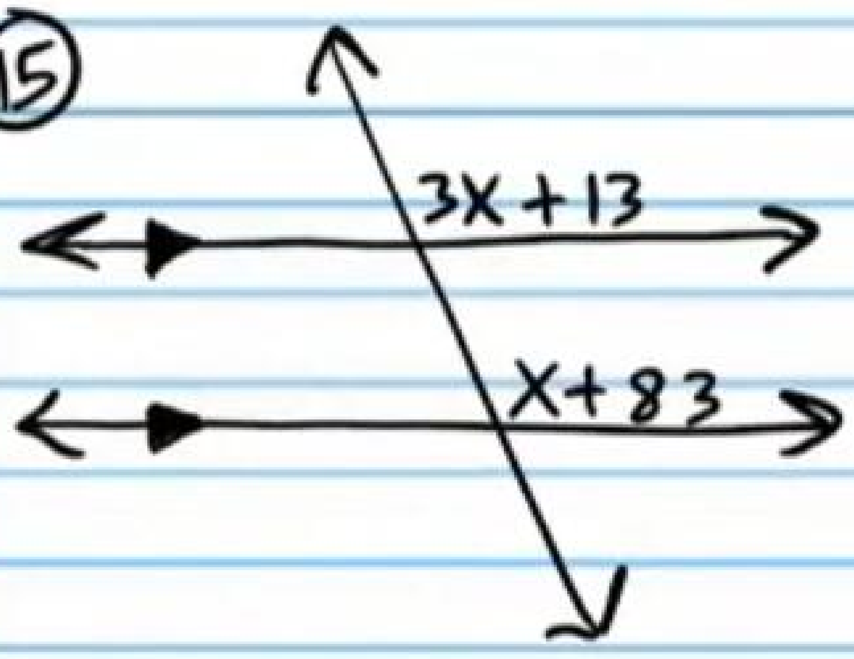
X=45
CAP

(14)



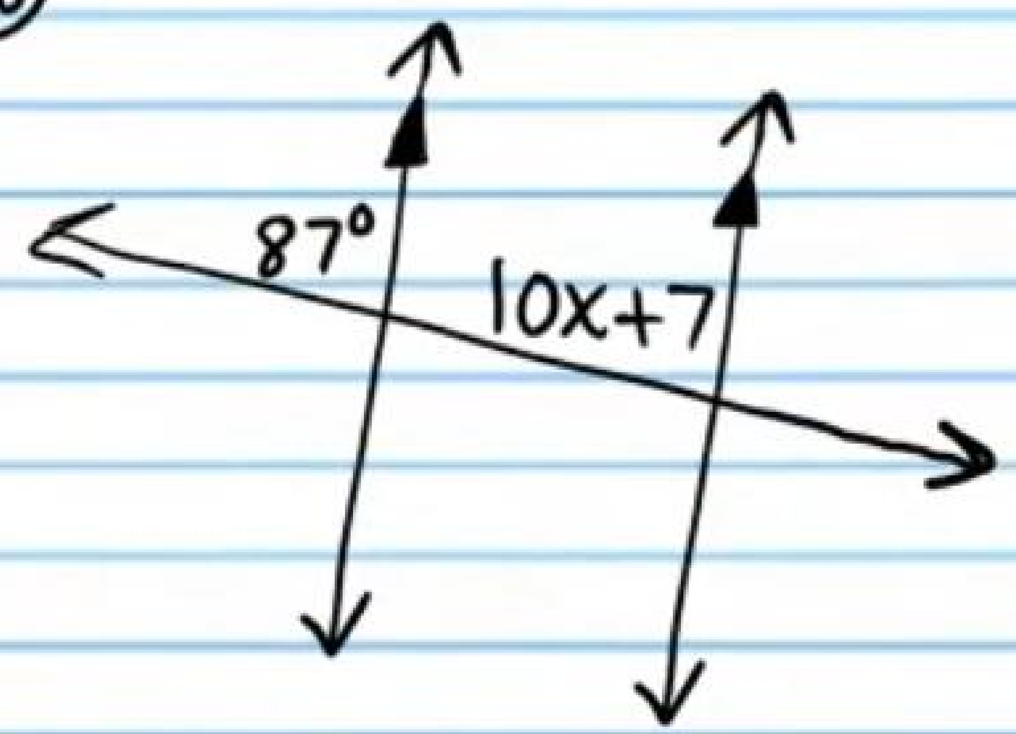
X=30
Linear Pair Postulate

(15)



$X=35$
CAP

(16)

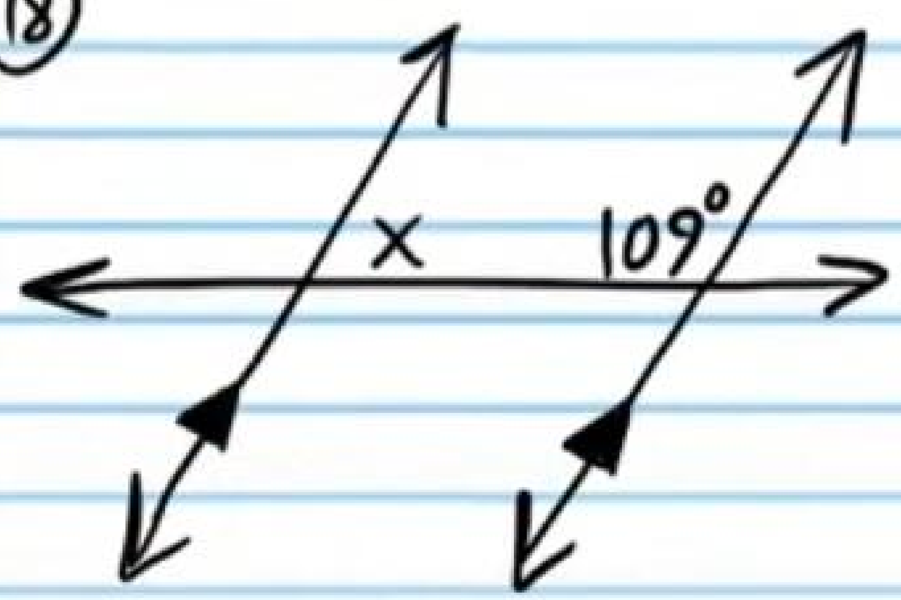


$X=8$
CAP

(17)

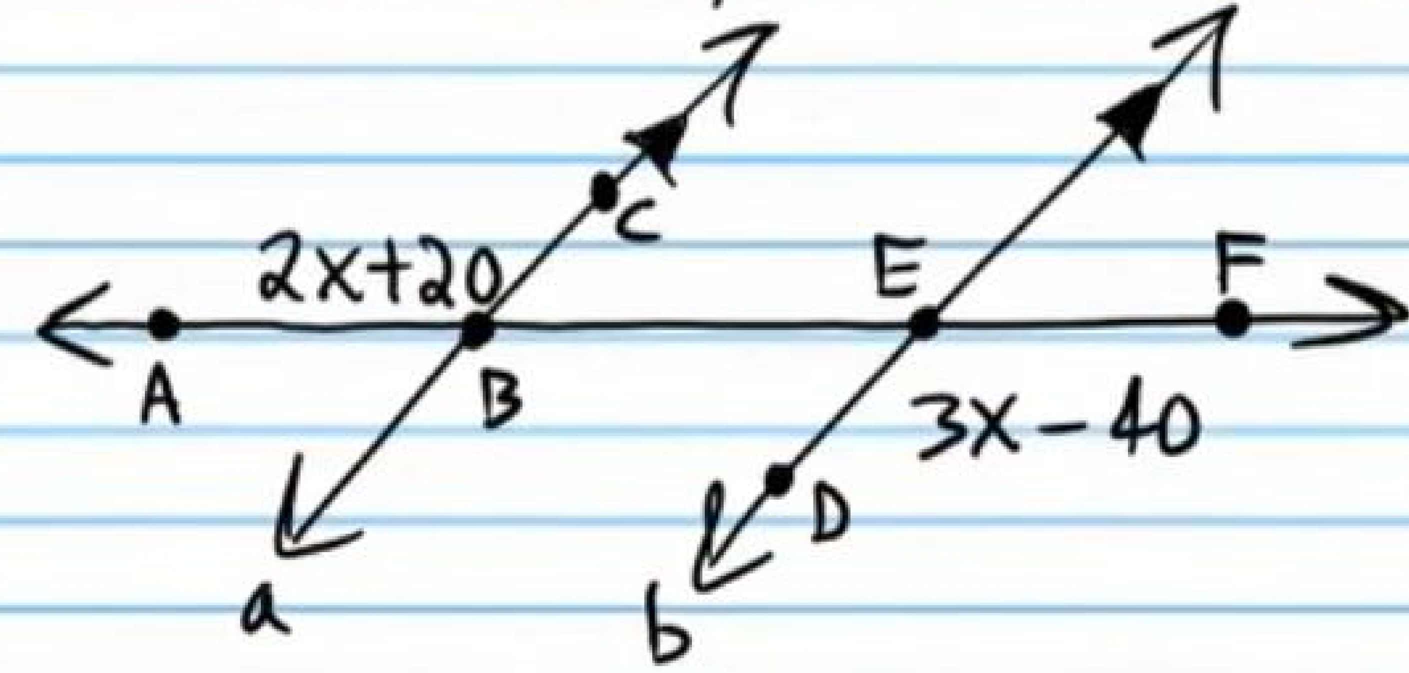
$X=57°$
Linear Pair Postulate and CAP.

(18)



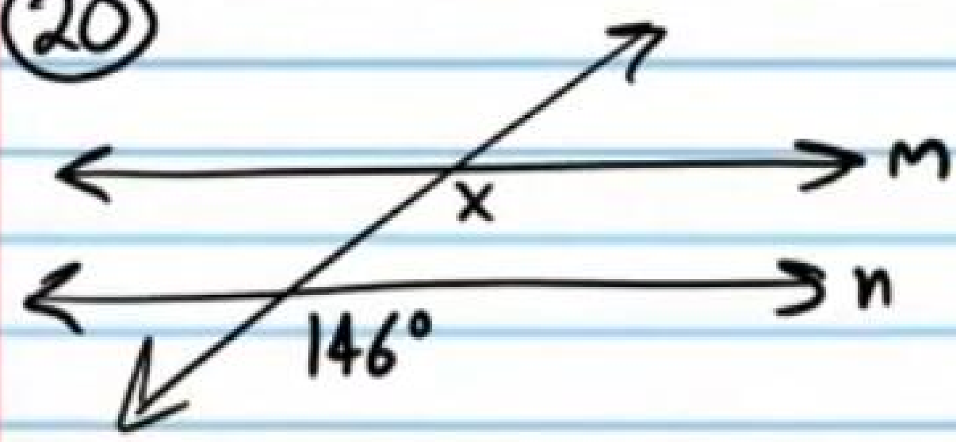
$X=71°$
CIAT

(19) Use a two-column proof to show that if lines a and b below are parallel, then $x=60$. The proof will be much faster and easier if you use one of the theorems presented in today's class.



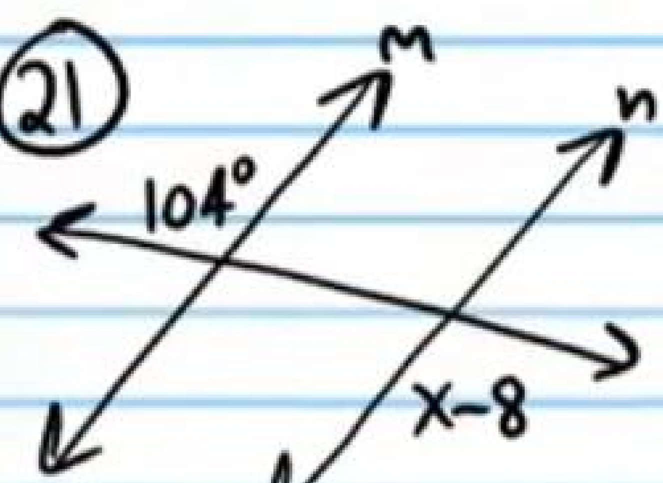
Find x such that lines m and n are parallel. State the postulate or theorem used.

(20)



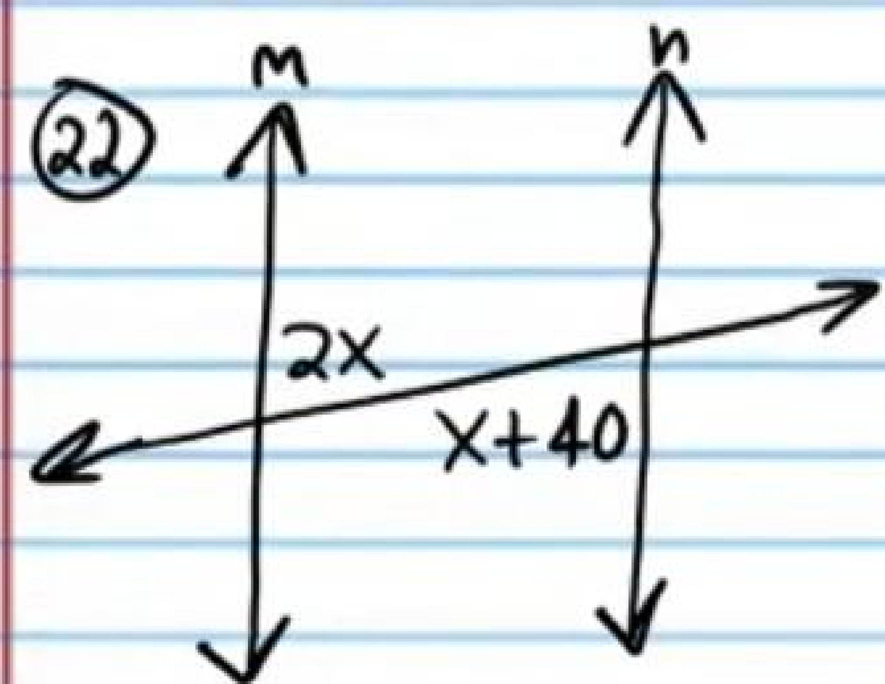
$X=146°$
CACP

(21)



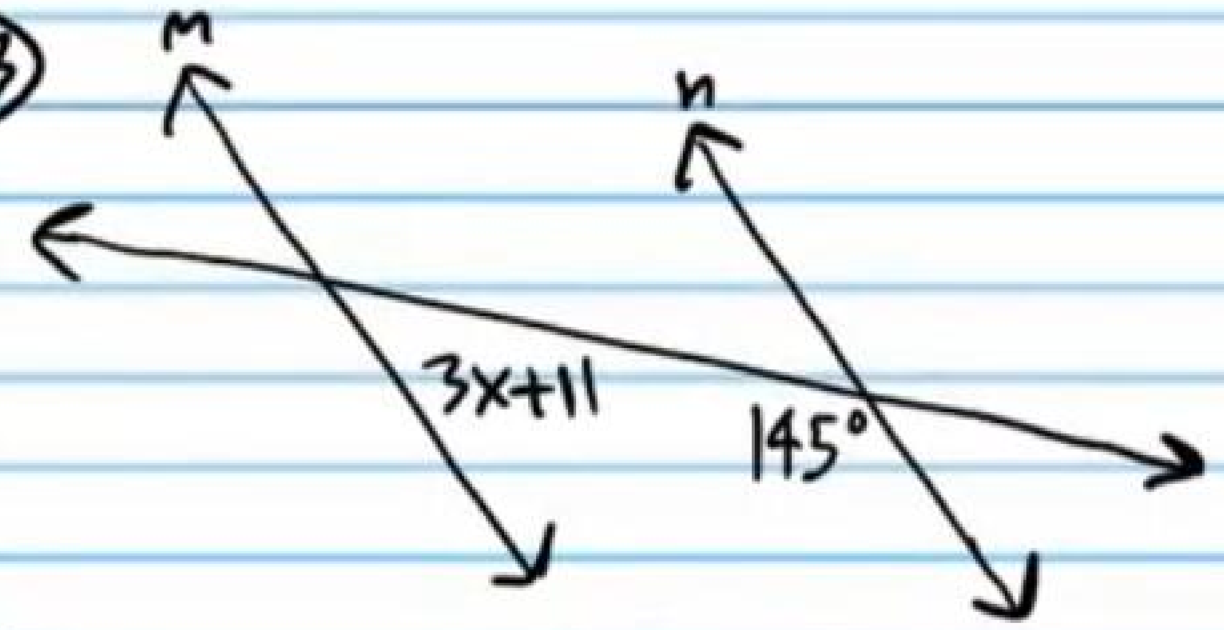
$X=112$
AEACT

(22)



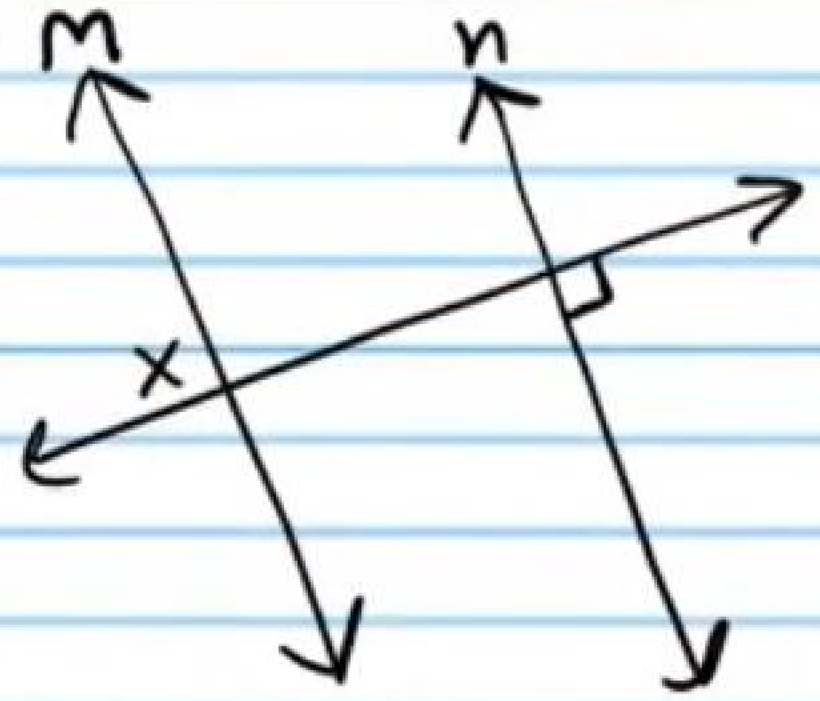
$X=40$
AIACT

(23)



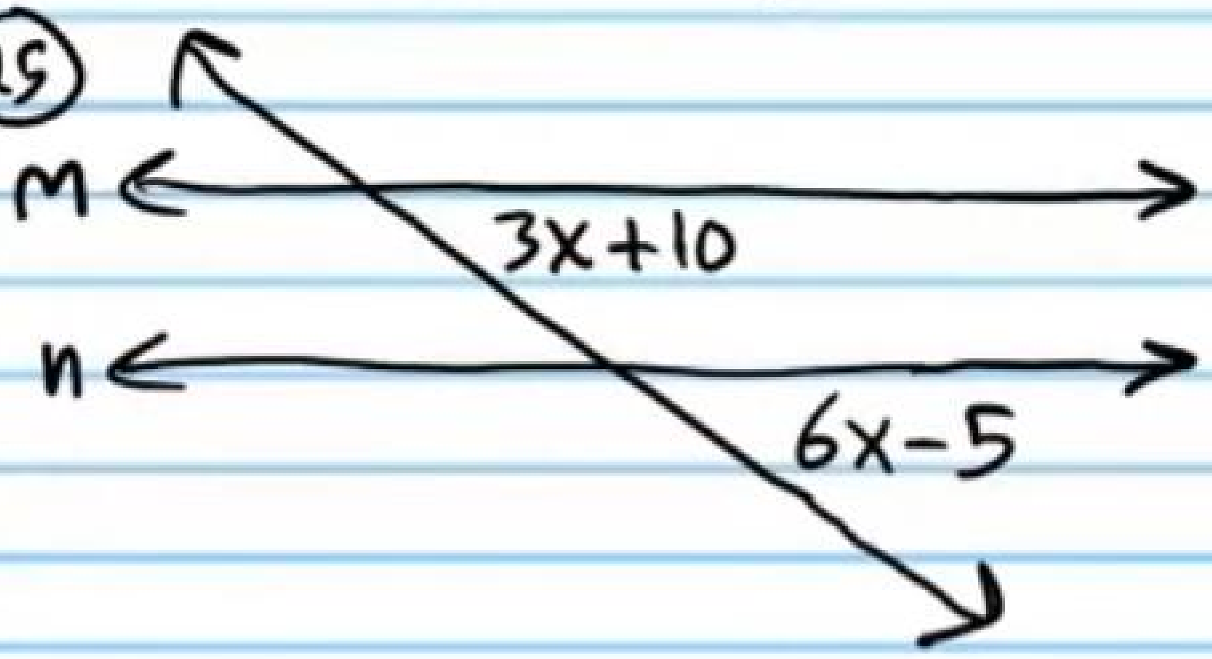
$\boxed{\begin{array}{c} x=8 \\ \text{CIACT} \end{array}}$

(24)



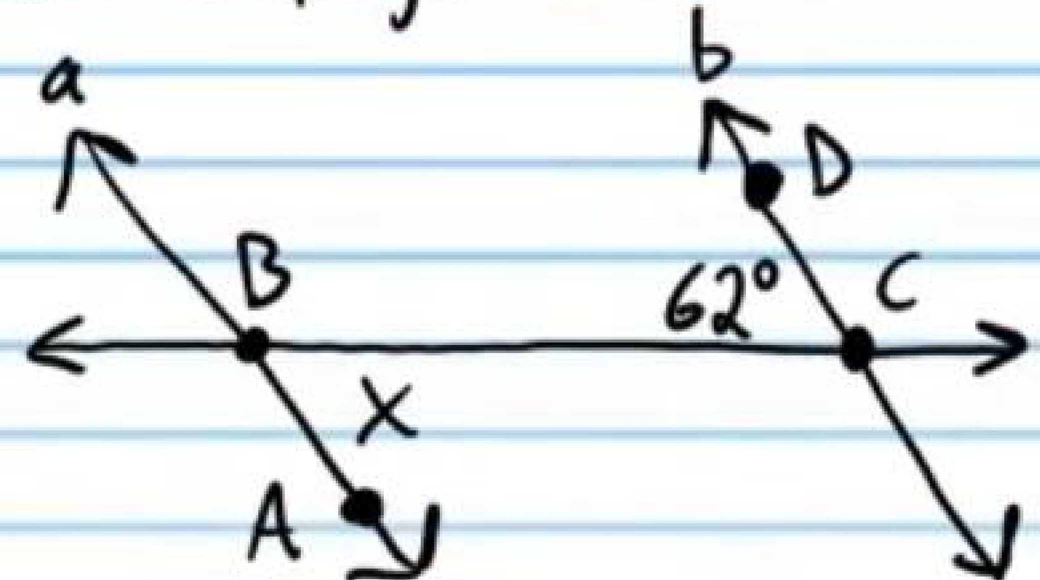
$\boxed{\begin{array}{c} x=90° \\ \text{AEACT} \end{array}}$

(25)



$\boxed{\begin{array}{c} x=5 \\ \text{CACP} \end{array}}$

(26) Use a two-column proof to show that if $x=62°$, then lines a and b below are parallel. The proof will be much faster and easier if you use one of the theorems we presented in today's class. The answer is on a different page.



Answers for Proofs

(19)

Statement	Reason
I) a ∥ b	Given.
II) $m\angle ABC = 2x+20$ and $m\angle DEF = 3x-40$.	Given (diagram)
III) $m\angle ABC = m\angle DEF$	Alt. Ext. ∠'s Thm.
IV) $2x+20 = 3x-40$	Subs. Prop of =.
V) $20 = x-40$	Subt. Prop of =.
VI) $60 = x$	Add Prop of =.
VII) $x = 60$	Symm. Prop of =.

(26)

Statement	Reason
I) $m\angle BCD = 62°$	Given.
II) $x = 62$	Given.
III) a ∥ b	Alt. Int. ∠'s Conv. Thm.